

# Ledger Entropy: Does an Entropy of the Log Exist?

Ledger Entropy Seminar

July 16, 2026

**Non-normative.** This note is an exploratory seminar work product. It states no requirement, binds no design, and amends no part of the specification or the Constitution. A negative result is a successful outcome.

## 1 The Question

The events of a ledger are not always fully verified. The loose statement “each event carries a probability of being accurate” is not crisp. We make it crisp by a restriction: every event under study records an *observed value*, and observed = true + Gaussian noise. Under that restriction, does a *ledger entropy* make sense? If entropy is the wrong functional, which functional is right? We define what kind of object such an entropy would be, name three candidates, and test each. The verdict is a qualified yes: a single absolute entropy of a ledger does not make sense, but a well-posed relative functional does.

## 2 Setting

Fix a valuation time and one conservation triple (unit, coordinate, agreement). Ten objects, in dependency order.

1. **Log**  $L$  — a finite append-only sequence of admitted events, ordered by admission.
2. **Event**  $e_i$  — one admitted record appended to the log at position  $i$ .
3. **Observed-value event** — an event carrying a numerical reading  $y_i \in \mathbb{R}^{d_i}$  together with an accuracy  $p_i > 0$ .
4. **Noise model** — the assumption  $y_i = A_i x + \varepsilon_i$ , with  $\varepsilon_i \sim \mathcal{N}(0, R_i)$ ,  $R_i = p_i^{-1} I$ , independent across events;  $A_i$  is a fixed observation map. Accuracy is precision:  $p_i$  large means  $R_i$  small.
5. **Candidate-log configuration** — a finite log together with its fixed noise model and prior; the complete data on which any functional is evaluated.
6. **True state**  $x \in \mathbb{R}^n$  — the unobserved vector of which the events are noisy readings.
7. **The fold** — the measurable map fold sending a configuration to its posterior parameters  $(\mu, \Sigma)$ , measurable in the readings  $(y_i)$ .
8. **The posterior** — the Gaussian law  $\mathcal{N}(\mu, \Sigma)$  on the true state induced by prior and noise model via fold.
9. **Attestation** — a refinement of an observed-value event that raises its accuracy  $p_i$  (shrinks its noise), leaving  $A_i$  fixed.
10. **Constraint subspace**  $V \subseteq \mathbb{R}^n$  — the affine subspace  $\{x : Cx = c\}$ , rank  $C = r$ , on which conserved quantities are fixed; the posterior is supported on  $V$ , so its ambient covariance  $\Sigma$  is rank-deficient.

**Notation.**  $L$  log,  $e_i$  event,  $y_i$  reading,  $p_i$  accuracy/precision,  $A_i$  observation map,  $R_i$  noise covariance,  $x$  true state,  $(\mu, \Sigma)$  posterior mean and covariance,  $V$  constraint subspace,  $H(\cdot)$  differential entropy,  $D(\cdot \| \cdot)$  relative entropy (Kullback–Leibler). No symbol is introduced elsewhere.

### 3 Candidate Definitions

What is random is not the state — it is a fixed unknown — but our *belief* about it, the posterior. Any ledger entropy is a functional of that posterior. Three candidates.

- E1** *Absolute differential entropy of the state posterior:*  $H = \frac{1}{2} \log \det(2\pi e \Sigma)$ .
- E2** *Hybrid mixture total.* Keep also the discrete layer: a configuration  $K$  records which candidate log is true, with weights  $w_k$  built from the accuracies  $p_i$ . The posterior is the mixture  $\pi = \sum_k w_k \mathcal{N}(\mu_k, \Sigma_k)$ , and E2 is its total entropy, combining discrete bits and continuous nats. This is the literal reading of “probability of being accurate.”
- E3** *Relative entropy / invariant information gain:*  $D(\text{posterior} \parallel \text{reference})$ , equal on the observation channel to the mutual information  $I(x; y)$ , with closed form  $\frac{1}{2} \log \det(I + \Sigma A^\top R^{-1} A)$ .

|           | well-posed     | invariant | compositional  | attest.-mono. | conserv.-compat. | computable |
|-----------|----------------|-----------|----------------|---------------|------------------|------------|
| <b>E1</b> | ×              | ×         | ✓              | ✓             | ×                | ✓          |
| <b>E2</b> | ×              | ×         | ◦ <sup>a</sup> | ✓             | ×                | ×          |
| <b>E3</b> | ◦ <sup>b</sup> | ✓         | ✓              | ✓             | ✓                | ✓          |

✓ holds unconditionally; × fails, with a counterexample in Results; ◦ holds under a stated side-condition. <sup>a</sup> the joint chain-rule total is additive over independent sub-logs, but the observable marginal does not factor. <sup>b</sup> well-posed once a reference is supplied; the ledger supplies it (see Verdict and Declaration).

The two E1 failures are *independent*: invariance fails for units, conservation fails for degeneracy, and repairing one does not repair the other (Prop. 3, Prop. 2, and its remark). E2 inherits both and adds a third, sharper one. E3 fails nothing structural; its only ◦ is the reference, which the ledger itself resolves.

### 4 Results

Throughout, the prior is Gaussian and proper on the free directions of  $V$ ; observations are linear-Gaussian.

**Proposition 1** (attestation reduces differential entropy). *An observation  $y = Ax + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, R)$ ,  $R \succ 0$ , sends the posterior  $\mathcal{N}(\mu, \Sigma) \mapsto \mathcal{N}(\mu', \Sigma')$  with  $\Sigma'^{-1} = \Sigma^{-1} + A^\top R^{-1} A$ , hence  $\Sigma' \preceq \Sigma$  and  $H' \leq H$ . The drop equals the information gained,  $H - H' = \frac{1}{2} \log \det(I + \Sigma A^\top R^{-1} A) = I(x; y) \geq 0$ , with equality iff  $A$  loads on no non-degenerate direction.*

*Proof.* The information form is standard conjugacy. Then  $\Sigma - \Sigma' = \Sigma A^\top (A \Sigma A^\top + R)^{-1} A \Sigma \succeq 0$ , of the form  $M^\top A M$  with  $A \succ 0$ ; determinant is monotone under the Loewner order on positive definite matrices, so  $\det \Sigma' \leq \det \Sigma$  and  $H' \leq H$ . The identity  $\det \Sigma / \det \Sigma' = \det(I + \Sigma A^\top R^{-1} A)$  follows from  $\Sigma'^{-1} = \Sigma^{-1}(I + \Sigma A^\top R^{-1} A)$ ; it is  $> 1$  iff  $A \neq 0$  on a free direction.  $\square$

**Proposition 2** (conservation forbids an absolute entropy). *Exact conservation  $Cx = c$ ,  $\text{rank } C = r$ , confines the posterior to the affine subspace  $V$ , which is Lebesgue-null in  $\mathbb{R}^n$ . Then  $C\Sigma = 0$ ,  $\det \Sigma = 0$ , and the ambient  $H = \frac{1}{2} \log((2\pi e)^n \det \Sigma) = -\infty$ . On  $V$  with its  $(n - r)$ -dimensional Lebesgue measure,  $H_V = \frac{1}{2} \log((2\pi e)^{n-r} \det_+ \Sigma)$  is finite, where  $\det_+$  is the product of the nonzero eigenvalues.*

*Proof.*  $\text{Var}(Cx) = C\Sigma C^\top = 0$  gives  $C\Sigma = 0$ , so  $\Sigma$  is singular and the ambient formula is  $-\infty$ . Restricting to an orthonormal basis of the tangent space of  $V$  replaces  $\det \Sigma$  by  $\det_+ \Sigma$ , which is finite.  $\square$

*Remark 1* (the two obstructions are independent).  $H_V$  cures degeneracy but not units: under a basis change  $B \mapsto BM$  of  $V$ ,  $H_V \mapsto H_V + \log |\det M|$ . Fixing the support does not fix the scale.

**Proposition 3** (differential entropy is not reparameterization-invariant). *For an invertible smooth change of coordinates  $y = g(x)$ ,  $H(y) = H(x) + \mathbb{E}[\log |\det J_g(x)|]$ ; for a linear rescale  $y = Bx$ ,  $H(Bx) = H(x) + \log |\det B|$ .*

*Proof.* Change of variables in the density,  $p_y(y) = p_x(g^{-1}(y)) |\det J_{g^{-1}}|$ , substituted into  $-\int p \log p$  (Cover–Thomas, ch. 8).  $\square$

Consequence: recording the free coordinates in cents rather than dollars ( $B = 100 I_{n-r}$ ) shifts  $H$  by  $(n-r) \log 100$ . The absolute number, and even its sign, is a unit artefact. Only differences, relative entropy, and mutual information survive — which is exactly the closed form of Prop. 1, invariant because the unit Jacobians cancel in the density ratio, and finite on  $V$  because the singular directions of posterior and reference cancel there. This is E3.

**Chain-rule identity (E2), and its unit caveat.** For the discrete configuration  $K$  and state  $X$ ,  $H(K, X) = H(K) + \sum_k P(k) H(X | K = k)$ ; this is an exact identity. But  $H(K)$  is dimensionless Shannon (bits,  $\geq 0$ , invariant) while  $H(X | K = k)$  is differential (carries units, sign-indefinite, non-invariant). Their sum adds bits to nats-with-units and inherits the reparameterization dependence of Prop. 3. This is the *sharpest* obstruction to E2: it ships only with a declared quantization scale  $\delta$  (using  $H_\delta(X) \approx H(X) - (n-r) \log \delta$  throughout), or by keeping the pair  $(H(K), H(X | K))$  unsummed, or by going relative on both — which is again E3.

**Second law (Heuristic).** The claim “entropy is non-decreasing between attestations and drops at each” is *false as stated*. With a deterministic fold and a static latent state, conservation-preserving moves push measure forward with  $|\det| = 1$ , so  $H$  is constant between attestations and falls only at them (Prop. 1); an append-only observation ledger has entropy that only ever ratchets *down*, the opposite of a thermodynamic second law. A genuine non-decrease requires an injected diffusion prior  $Q \succeq 0$  ( $\Sigma \mapsto F \Sigma F^\top + Q$ ), a modelling choice not forced by the framework. With it the picture is a sawtooth: rise by  $Q$  between, drop at each attestation. Status: heuristic, conditional on a declared  $Q$ .

**Mixture marginal (cited fact).** The marginal of the E2 mixture  $\sum_k w_k \mathcal{N}(\mu_k, \Sigma_k)$  has no closed-form differential entropy (Huber et al., 2008); only bounds and approximations exist. So the single E2 total is not closed-form computable, and summing its  $2^{|L|}$  configuration terms is exponential.

## 5 Worked Example

This instantiates Prop. 1 (attestation) and Prop. 3 (units). One instrument; wallets  $A, B$ , issuer  $I$ ; latent quantity the true price  $x$ . Accuracy is precision,  $p = 1/\text{variance}$ .

| event                                   | effect                       | reading       | noise                             |
|-----------------------------------------|------------------------------|---------------|-----------------------------------|
| $e_1$ issue 100 to $A$                  | $I : -100, A : +100$ (exact) | —             | none                              |
| $e_2$ observe price                     | valuation                    | $y_1 = 50.00$ | $p_1 = 100$ ( $\sigma_1 = 0.10$ ) |
| $e_3$ transfer $A \rightarrow B$ qty 40 | $A : -40, B : +40$ (exact)   | —             | none                              |
| $e_4$ attest $e_2$                      | valuation                    | $y_2 = 50.04$ | $p_2 = 400$ ( $\sigma_2 = 0.05$ ) |

*Chair ruling (ratified): quantities are exact; only values are noisy.* A move is an instruction, not a measurement. Conservation vanishes by paired-leg construction precisely because the two legs are

the same number posted with opposite signs. If a transfer quantity were a noisy observation, either both legs share the draw (the noise cancels in the wallet sum and the “noise” is illusory) or the legs are independent draws (conservation is violated in expectation, which the framework forbids). So after  $e_3$ ,  $A$  holds exactly 60 units and  $B$  exactly 40; all uncertainty lives on the price.

**Posterior, two independent ways.** Sequential update (flat prior;  $e_2$  sets  $\mathcal{N}(50.00, 0.01)$ ; gain  $K = 0.01/0.0125 = 0.8$ ): mean =  $50.00 + 0.8(0.04) = \mathbf{50.0320000000}$ , variance =  $0.2 \times 0.01 = \mathbf{0.0020000000}$ . Batch precision-weighting ( $p_1 = 100, p_2 = 400$ ): variance =  $1/500 = \mathbf{0.0020000000}$ , mean =  $0.002(100 \cdot 50.00 + 400 \cdot 50.04) = \mathbf{50.0320000000}$ . Identical to ten digits:  $x \sim \mathcal{N}(50.032, 0.002)$ .

**E1, two independent ways.**  $H = \frac{1}{2} \ln(2\pi e \sigma^2)$  (closed form) versus  $-\int f \ln f$  (numerical quadrature,  $\pm 12\sigma$ ):

| state       | variance | $H$ closed (nats) | $H$ quadrature (nats) |
|-------------|----------|-------------------|-----------------------|
| after $e_2$ | 0.01     | −0.883647         | —                     |
| after $e_4$ | 0.002    | − <b>1.688366</b> | − <b>1.688366</b>     |

The negative sign and the unit-dependence of these levels are exactly Prop. 3: read the price in cents and every  $H$  shifts by  $\ln 100$ . The *change*, however, is invariant. **E3:** the attestation deposits  $H - H' = \frac{1}{2} \ln(0.01/0.002) = \frac{1}{2} \ln 5 = \mathbf{0.804719}$  nats = 1.160964 bits =  $I(x; y_2 | y_1)$ , matching  $\frac{1}{2} \ln(1 + p_2 \sigma_{\text{before}}^2) = \frac{1}{2} \ln(1 + 400 \cdot 0.01) = \frac{1}{2} \ln 5$ ; and  $D(\text{posterior} \parallel \text{prior}) = 0.4559$  nats. The two marks agree ( $|\Delta| = 0.04$  versus  $\sigma_1 = 0.10$ , about  $0.4\sigma$ ): no attestation dispute. The operator’s reading is the per-position mark and its confidence interval —  $A$ ’s line  $60 \times 50.032 = 3001.9 \pm 2.7$ ,  $B$ ’s  $40 \times 50.032 = 2001.3 \pm 1.8$  — and the per-attestation gain in bits, both invariant; not  $H$  itself.

## 6 Verdict

*Signed, GROTHENDIECK, Chair.* The word “entropy” is the wrong word for a ledger, and the right functional is *relative entropy*, the invariant information gain E3. An absolute entropy of a ledger state (E1) fails two independent tests: it is not *invariant* — Prop. 3, dollars-to-cents moves the number and its sign — and it is not *conservation-compatible* — Prop. 2, the conservation law pins the state to a subspace where the ambient entropy is  $-\infty$ . Curing one does not cure the other. The hybrid total (E2), which takes the “probability of being accurate” literally, inherits both and adds a third and sharper defect: it sums Shannon bits to differential nats, quantities of different type, and has no closed form. What survives all six tests is E3: the relative entropy of the current posterior against a reference, equal to the information the event deposits about the truth. It is unit-invariant (the Jacobians cancel in a ratio); degeneracy-safe (posterior and reference share the constraint subspace, so singular directions cancel); additive over independent sub-logs, hence checkpoint-independent and order-insensitive; monotone under attestation; and cheap, needing only the marginal covariance on the coordinates an observation touches. The one thing E3 needs is a reference, and the ledger already keeps one: the *as-known-at* state before the event. So the honest object is not “the entropy of the ledger” but “the information this event added, measured against what was known before.” Entropy does not make sense as a level. Information gain makes sense as a difference. That difference is the functional the ledger admits.

## 7 Open Questions and Related Work

**Classification.** Under the restriction, this is a constrained Kalman / information filter over a Gaussian field — a linear-Gaussian estimation problem in an entropy costume. Every candidate is a scalarization of the posterior covariance  $\Sigma$ : E1 is  $\log \det \Sigma$  (D-optimality),  $\text{tr} \Sigma$  and  $\lambda_{\max}(\Sigma)$  its A- and E-optimality siblings. E3 is the Bayesian experimental-design criterion (Lindley), which turns the survivor into a policy: point the scarce attestation budget at the observation of largest expected gain.

**Related work** (contribution / bearing on this question). *Shannon* 1948 — coding rate of a source; fixes what this is not, the log is a record. *Jaynes* 1957 — the prior is max-entropy given the conservation moments. *Kalman* 1960 — the actual engine; the restricted ledger is an information filter. *Kullback–Leibler* 1951 — relative entropy, the invariant object E3. *Amari* — Fisher information  $\Sigma^{-1}$  as metric;  $D$  is its second-order form. *Baez–Fritz–Leinster* 2011 — entropy as information loss along a map; the fold is lossless replay, so the analogue is gain, not loss. *Fritz* 2020 — Markov categories; a second telling of the update only. *Pearl* — belief propagation, the feasible computation on the sparse coordinate graph. *Valiant* (PAC) — “probability an event is accurate” as a learnable reliability parameter, orthogonal to the functional. *Dempster–Shafer* — the imprecise-probability layer the Gaussian restriction deletes; needed back only if  $p_i$  were an interval. *Lindley* 1956 — Bayesian experimental design, the operational keep. *Prior art, not a source*: the audit-risk model  $AR = IR \times CR \times DR$  (AICPA; ISA/AU-C 200) is a heuristic product of subjective probabilities, with no derivation the ledger could reuse.

**Open.** (i) The diffusion prior  $Q$  that would make a second law non-vacuous — what, if anything, in the framework supplies it. (ii) A discrete-layer attestation calculus: E2 read as “which events are admitted” is a genuine object; its well-posed relative form is unsettled. (iii) Computability at scale: for  $\sim 10^6$  live coordinates a global log det is governed by the treewidth of the coordinate-dependency graph, which must be measured, not assumed; the local E3 gain is cheap regardless.

*Remark 2* (second telling; delete this box and nothing above changes). In the Kleisli category of the Giry monad, the noisy log is one arrow  $\Theta \rightarrow S$ : a tensor of Gaussian observation channels followed by the deterministic fold. The posterior is its Bayesian inverse; attestation is post-composition of a further channel. Information gain is a monoid homomorphism  $([Event], ++)\rightarrow(\mathbb{R}_{\geq 0}, +)$ , which is why the total of E3 is checkpoint-independent and order-insensitive by construction. No proof above rests on this paragraph.

## 8 Declaration

**Sessions.** Sessions 1–4 were used: nine independent Session-1 formalizations by distinct routes (analytic, symmetry, honesty gate, categorical, unifying object, rigor veto, architecture, classification, framing); a Session-2 synthesis into E1/E2/E3 against the six criteria; a Session-3 worked example; this Session-4 write-up.

**Chair rulings.** (1) *The toy decision*: quantities are exact, only values are noisy, on the conservation argument given in the Worked Example — ratified. (2) *The reference gap*: the reference E3 requires is *closed* for the incremental gain and remains a *conjecture* for any absolute total. The incremental reference is the previous as-known-at posterior, which the bitemporal doctrine already maintains — no new object is posited, so the per-event gain is well-posed. A cumulative total would need a canonical origin (the initial prior); that the max-entropy-given-conservation prior is that origin is natural but not forced, hence a conjecture, not a proposition.

**Labels.** The rigor constraints W1–W7 were enforced: every differential quantity names its reference; only invariants are headline numbers; all statements live on the constraint subspace of dimension  $n - r$ ; bits and nats are never summed without a declared  $\delta$ ; each object is named; the second-law claim names the diffusion prior it needs; the mixture marginal is invoked only with its no-closed-form citation. Only Propositions 1–3 carry proofs; the chain rule is an identity with a caveat; the second law is a heuristic; the mixture fact is cited. The E3 closed form is stated as a corollary of Proposition 1, not minted as a new theorem — the one label decision this chair made to stay within the approved set.

**Dissents, independence, confidence.** No dissents. The nine notes shared no cross-reading yet converged by distinct routes — either strong evidence or a common prior in the authors’ training; recorded as convergence, not proof. Confidence: high on the classification and the invariance argument; medium on global computability, which hinges on a treewidth that must be measured.